

Jessica Coulson

438-528-8526 | jessica.coulson@mail.mcgill.ca | [linkedin.com/in/jessicalcoulson](https://www.linkedin.com/in/jessicalcoulson)

EDUCATION

McGill University

Aug. 2022 – Present

B.Eng. in Bioengineering, Minor in Applied Artificial Intelligence

Montréal, QC

- CGPA: 3.96/4.00
- Dean's Honor List (2022-23, 2023-24), 1989 Polytechnique Memorial Scholarships (2023-24, 2024-25)

PROFESSIONAL EXPERIENCE

Research Intern

August 2025 - Present

Mila - Quebec AI Institute

Montréal, QC

- Integrate longitudinal brain imaging analysis with single-cell transcriptomics analysis to evaluate cross-modal disease signatures, under the supervision of Dr. Danilo Bzdok
- Apply Partial Least Square regression to assess correlation between brain imaging features and Alzheimer's markers

Signal Processing Intern

April 2025 - August 2025

Research Institute of the McGill University Healthcare Center (RI-MUHC)

Montréal, QC

- Applied signal processing and analysis techniques in the BIAPT lab under Dr. Stefanie Blain-Moraes on physiological signals of dancers and residents with advanced dementia, investigating interpersonal synchrony
- Adapted and implemented synchrony algorithms, addressing unique physiological patterns of residents to enable data-driven evaluation of embodied memory progression

Research Intern

May 2024 – August 2024

McGill University

Montréal, QC

- Investigated the mechanisms of human aortic valve calcification under Dr. Marta Cerruti in the Biointerface Lab (NSERC USRA-funded)
- Proactively designed and led novel experimental protocol, resulting in first-ever documented *in vitro* synthesis of target mineral phase under physiological conditions
- Conducted independent data analysis and presented findings at the McGill SURE Poster Session

LEADERSHIP EXPERIENCE

Co-President

June 2024 – June 2025

Promoting Opportunities for Women in Engineering (POWE)

Montréal, QC

- Recruited a team of 10 executives and 35+ coordinators, acting as the primary spokesperson on school boards
- Organized large-scale conferences, panels, and networking events for 200+ students, facilitating professional development and engagement with industry leaders
- Launched the MiMi Aung Research and Innovation Competition, attracting 40+ student projects

TECHNICAL SKILLS

Programming Languages: Python, Java, R, SQL

Tools: Git/GitHub, VS Code, IntelliJ, AWS, Jupyter Notebooks, Linux/Bash

Libraries: scikit-learn, pandas, NumPy, matplotlib, PyTorch

Additional: MATLAB, LaTeX, Autodesk Fusion 360, JMP (SAS), Figma, Microsoft Office Suite

COMPETITIONS AND AWARDS

People's Choice in MAIS Hackathon

November 2025

Built SpeakEZ, an AI-driven platform to prep students for interviews by tracking body language and filler words.

1st Place in BOLT Case Competition x Hackathon

January 2025

Pitched and designed Kaleido, an AI-driven solution to streamline consultant-to-project matching for Levio

2nd Place & Audience Favorite in TechIdea Pitch Competition

March 2023

Presented a sustainable fabric solution in the ideation stage

1st Place in BRIDGE Case Competition

January 2023

Pitched a novel, sustainable process to create fabric from soy whey; recognized by Frank and Oak CFO